

Adaptive Kato Transport of Overlap-Continued Circuit Projectors: A Preregistered Width Comparison

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¹*Class-A fermionic adaptive-circuit project*
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We specify a geometric flux-continuation experiment on saved pure Gaussian states produced by the monitored class-A circuit. At each flux, a fixed-rank spectral subspace of the state-derived flattened parent is selected by maximum overlap with the preceding projector. Its frame is then transported with the Kato connection rather than a physical-time Schrödinger ramp. Adaptive classical RK4 uses projector step doubling, while the selected-excluded spectral gap is retained as a diagnostic. The campaign compares 20×24 and 24×24 endpoint ensembles, separately reporting raw clockwise and counterclockwise wall-charge transfer. Numerical validity is gated, but charge quantization and agreement with earlier finite-grid continuations are measured outcomes. This document fixes the theory, algorithm, diagnostics, and resume contract before production results are inspected.

I. QUESTION AND SCOPE

The equilibrium Laughlin argument identifies a Chern phase by transporting an occupied spectral subspace around one boundary-flux quantum. A monitored circuit supplies several inequivalent flux-dependent objects. A physical online circuit ramp includes Born noise and feedback, and a finite-time Schrödinger ramp of a state-derived Hamiltonian includes Landau-Zener mixing. Neither is identical to geometric continuation of a spectral branch. The present experiment isolates the latter operation.

The input is a completed monitored endpoint state. No circuit trajectory is rerun, no tangent cocycle is constructed, and no physical ramp time is introduced. The output is the direct half-cylinder charge of a projector transported continuously in flux. This is the closest circuit-state analogue of the overlap-continued flattened-Hamiltonian benchmark, but it does not make quantization automatic: the endpoint family may lack a clean isolated branch, the finite cylinder contains two walls, and previous-overlap selection can become ambiguous near a degeneracy.

II. PURE GAUSSIAN STATES AND THEIR PROJECTORS

A number-conserving pure Gaussian state with occupied frame $F \in \mathbb{C}^{N \times r}$ satisfies

$$F^\dagger F = \mathbf{1}_r, \quad P = FF^\dagger, \quad P^2 = P = P^\dagger. \quad (1)$$

The frame basis is not physical: F and FU represent the same state for every $U \in U(r)$. The rank- r projector P is the physical one-particle correlation matrix. Regional particle numbers therefore follow directly from its diagonal,

$$N_A = \text{tr}(R_A P), \quad (2)$$

where R_A projects onto a spatial region.

For an endpoint projector P_0 , define the state-derived flattened parent

$$H_0 = \mathbf{1}_N - 2P_0. \quad (3)$$

It has occupied eigenvalue -1 and empty eigenvalue $+1$ at zero twist. Threading a uniform boundary flux in the periodic y direction gives

$$\tilde{H}_{rs}(\phi) = (H_0)_{rs} \exp \left[i\phi \frac{d_y(r, s)}{N_y} \right], \quad (4)$$

$$H(\phi) = \frac{\tilde{H}(\phi) + \tilde{H}^\dagger(\phi)}{2}, \quad (5)$$

with minimum-image displacement d_y . The derivative is analytic,

$$\partial_\phi \tilde{H}_{rs} = i \frac{d_y(r, s)}{N_y} \tilde{H}_{rs}, \quad (6)$$

$$\partial_\phi H = \frac{\partial_\phi \tilde{H} + (\partial_\phi \tilde{H})^\dagger}{2}. \quad (7)$$

III. THREE DISTINCT FLUX-DEPENDENT PROJECTORS

The notation distinguishes three operations that can otherwise look deceptively similar.

The *dynamical projector* $P_{\text{dyn}}(t)$ solves a physical-time equation such as

$$i\partial_t F_{\text{dyn}} = H[\phi(t)]F_{\text{dyn}}. \quad (8)$$

Its result depends on ramp time and can mix instantaneous branches at an avoided crossing.

The *instantaneous projector* $P_{\text{inst}}(\phi)$ is rebuilt independently at every flux by filling the r lowest eigenvectors of $H(\phi)$. It has no memory of how the flux point was reached. In a finite system it normally closes after 2π by refilling according to energy order.

The *continued projector* $P_{\text{cont}}(\phi)$ remembers the previous spectral subspace. At a numerical stage with eigenpairs $\{E_n, |w_n\rangle\}$, every eigenvector receives the gauge-invariant weight

$$\omega_n = \langle w_n | P_{\text{previous}} | w_n \rangle. \quad (9)$$

The fixed-rank set I_{cont} consists of the r largest weights. Only exact numerical ties are resolved by increasing energy, reproducing the legacy static selector. The selected spectral projector is

$$P_{\text{spec}}(\phi) = \sum_{i \in I_{\text{cont}}} |i\rangle\langle i|. \quad (10)$$

If the branch is smooth and resolved, Kato transport makes $P_{\text{cont}} = P_{\text{spec}}$ up to integration error. We keep the two names because their discrepancy is a primary numerical diagnostic.

IV. KATO PARALLEL TRANSPORT

A. Derivation from $P^2 = P$

Differentiating $P^2 = P$ gives

$$(\partial_\phi P)P + P(\partial_\phi P) = \partial_\phi P, \quad P(\partial_\phi P)P = 0. \quad (11)$$

Thus $\partial_\phi P$ is off diagonal between the selected subspace and its complement. Define the anti-Hermitian Kato generator

$$K(\phi) = [\partial_\phi P_{\text{spec}}, P_{\text{spec}}], \quad K^\dagger = -K. \quad (12)$$

The frame equation is

$$\partial_\phi F_{\text{cont}} = K(\phi)F_{\text{cont}}. \quad (13)$$

It induces the basis-independent projector equation

$$\partial_\phi P_{\text{cont}} = [[\partial_\phi P_{\text{spec}}, P_{\text{spec}}], P_{\text{cont}}]. \quad (14)$$

Indeed, $\partial_\phi(FF^\dagger) = KP - PK = [K, P]$; when $P = P_{\text{spec}}$, Eq. (11) gives $[K, P] = \partial_\phi P$. The connection fixes a convenient parallel gauge for the frame but transports the projector, which is the physical object.

B. Analytic spectral derivative

Away from an unresolved selected–excluded degeneracy, first-order spectral perturbation theory gives

$$\partial_\phi P_{\text{spec}} = \sum_{\substack{i \in I_{\text{cont}} \\ a \notin I_{\text{cont}}}} \frac{|a\rangle\langle a|(\partial_\phi H)|i\rangle\langle i|}{E_i - E_a} + \text{H.c.} \quad (15)$$

The selected–excluded gap

$$\delta_{\text{sel}} = \min_{\substack{i \in I_{\text{cont}} \\ a \notin I_{\text{cont}}}} |E_i - E_a| \quad (16)$$

controls the norm that Eq. (15) can acquire. It is not, by itself, an integration-error estimate. Matrix elements of $\partial_\phi H$, rotation distributed among many modes, and branch changes also matter. For that reason δ_{sel} is recorded but does not set the flux step directly.

V. ADAPTIVE RK4 ALGORITHM

The regulated observation grid is

$$\phi_j^{(\sigma)} = -\sigma 10^{-7} + \sigma \frac{2\pi j}{128}, \quad j = 0, \dots, 128, \quad \sigma = \pm 1. \quad (17)$$

The signed 10^{-7} displacement starts on a definite side of the ambiguous finite-size crossing at the nominal origin. It does not alter the Hamiltonian away from that negligible shift and is not a quantization criterion.

For each RK4 stage, the code constructs H and its analytic derivative, diagonalizes H , performs the previous-projector selection, constructs Eq. (15), and applies the generator in Eq. (13). The same proposed step is evaluated once at size h and twice at size $h/2$. After symmetric polar orthonormalization of the two candidates, the local projector discrepancy is

$$e = \frac{\|P_{h/2, h/2} - P_h\|_F}{\sqrt{r}}. \quad (18)$$

A step is accepted only for $e \leq 10^{-8}$. The next proposal is

$$h_{\text{new}} = 0.9h \left(\frac{10^{-8}}{e} \right)^{1/5}, \quad 0.2h \leq h_{\text{new}} \leq 2h. \quad (19)$$

No step may cross an observation point. Failure is diagnostic if the proposal falls below $1/4096$ of one observation interval. This rule catches an unresolved branch or singular derivative instead of silently stepping through it.

After every accepted step, the frame is stabilized by the symmetric polar map

$$F \longmapsto F(F^\dagger F)^{-1/2}. \quad (20)$$

The result is *not* reprojected onto P_{spec} . Reprojection would hide accumulated transport error and could convert a branch-selection discontinuity into an apparently exact path. The direct projector mismatch $\|P_{\text{cont}} - P_{\text{spec}}\|_F / \sqrt{r}$ remains observable and must stay below 10^{-7} .

VI. CAMPAIGN AND SAVED OBSERVABLES

The inputs are verified complex128 endpoint frames from two completed raster- y monitored campaigns. Both used pure initialization, perfect correction, no postselection, $n_{\text{shell}} = 1$, and $(\alpha_1, \alpha_2) = (1, 30)$ for 48 cycles. The width comparison is

size	walls	split	selected IDs
20×24	$x = 5, 15$	$x = 10$	$0, 4, \dots, 96$
24×24	$x = 6, 18$	$x = 12$	$0, 4, \dots, 96$

Soft and hard walls are both included. Each of the 100 endpoint states is continued clockwise and counterclockwise, yielding 200 paths. The two sizes are independent monitored ensembles; the two directions for a fixed endpoint are paired deterministic continuations.

At all 129 points we save ϕ , N_L , N_R , total charge,

$$q_x = \frac{(N_R - N_R^{(0)}) - (N_L - N_L^{(0)})}{2}, \quad (21)$$

the x -resolved density, transported–selected projector mismatch, minimum principal overlap, selected-weight floor and margin, δ_{sel} , generator norm, adaptive error, and accepted/rejected step counts. Raw CW and CCW distributions are reported separately. The analysis compares each path to its saved 64-interval static continuation and also to the 128-interval static continuation at 20×24 . Every change in the descriptive $|q_x| > 0.5$ classification is listed. Fractions above 0.5 and 0.75 are descriptive outcomes, not gates.

Each path writes one atomic compressed NPZ and one completion JSON written last. The completion binds the task, full source endpoint identity, configuration hash, source hashes, result bytes, and SHA-256. A rolling uncompressed frame checkpoint is written every eight observation intervals as one stable NPZ/JSON pair. It

contains the exact transported frame, completed interval, next adaptive proposal, accumulated diagnostics, and all partial histories. Restart verifies and skips complete pairs or continues a matching checkpoint; corrupt or partial pairs are rejected.

VII. NUMERICAL GATES AND INTERPRETATION

The preregistered numerical gates are: source-frame Gram residual below 10^{-8} ; post-polar orthonormality residual below 10^{-10} ; total-charge residual below 10^{-9} ; transported–selected projector mismatch below 10^{-7} ; finite spectral and adaptive diagnostics; and step-doubling error not exceeding 10^{-8} . Sample 0 for both sizes, walls, and directions is repeated at tolerance 2.5×10^{-9} , with full-path and endpoint raw q_x differences required below 10^{-4} .

Kato continuation removes physical ramp time and hence removes finite-rate Landau–Zener mixing from the definition of the path. It cannot force charge quantization. A true ambiguity can remain if the selected–excluded gap closes, if the overlap margin vanishes, or if a finite-system continued branch does not carry a full unit of wall charge. Quantization, agreement with the coarse-grid classification, and width improvement are therefore reported only after all 200 path pairs verify. No result claim is made in this methods note.

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